
Education

University of Utah*B.S. Computer Science***Salt Lake City, UT***August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies*Associate Software Engineer***Dallas, TX***June 2025 – Present*

- Built containerized microservices for high-frequency data ingestion, processing, and storage using Podman and serverless architecture on defense sensor streams.
- Developed signal processing algorithms in Python and C++ on embedded systems, meeting strict latency and resource constraints in collaboration with hardware teams.
- Automated CI/CD pipelines with static code analysis and integration testing, reducing deployment friction across engineering teams.

Doxy.me*Software Engineer***Charleston, SC***August 2024 – May 2025*

- Owned AWS cloud infrastructure for a HIPAA-regulated healthcare platform, managing container orchestration on ECS and Fargate, SageMaker inference clusters, and Lambda functions while reducing deployment risk through infrastructure-as-code automation and rigorous environment parity.
- Improved platform reliability by defining observability standards across distributed Python and TypeScript services, including structured logging, alerting thresholds, and incident response runbooks, reducing mean time to resolution for production failures.
- Built self-serve CI/CD primitives and shared deployment tooling that allowed product engineers to provision and ship new features to the healthcare platform independently, reducing infrastructure wait time and increasing weekly release cadence.

University of Utah*Undergraduate Research Assistant – FuTURES Lab***Salt Lake City, UT***May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
 - Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.
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Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.